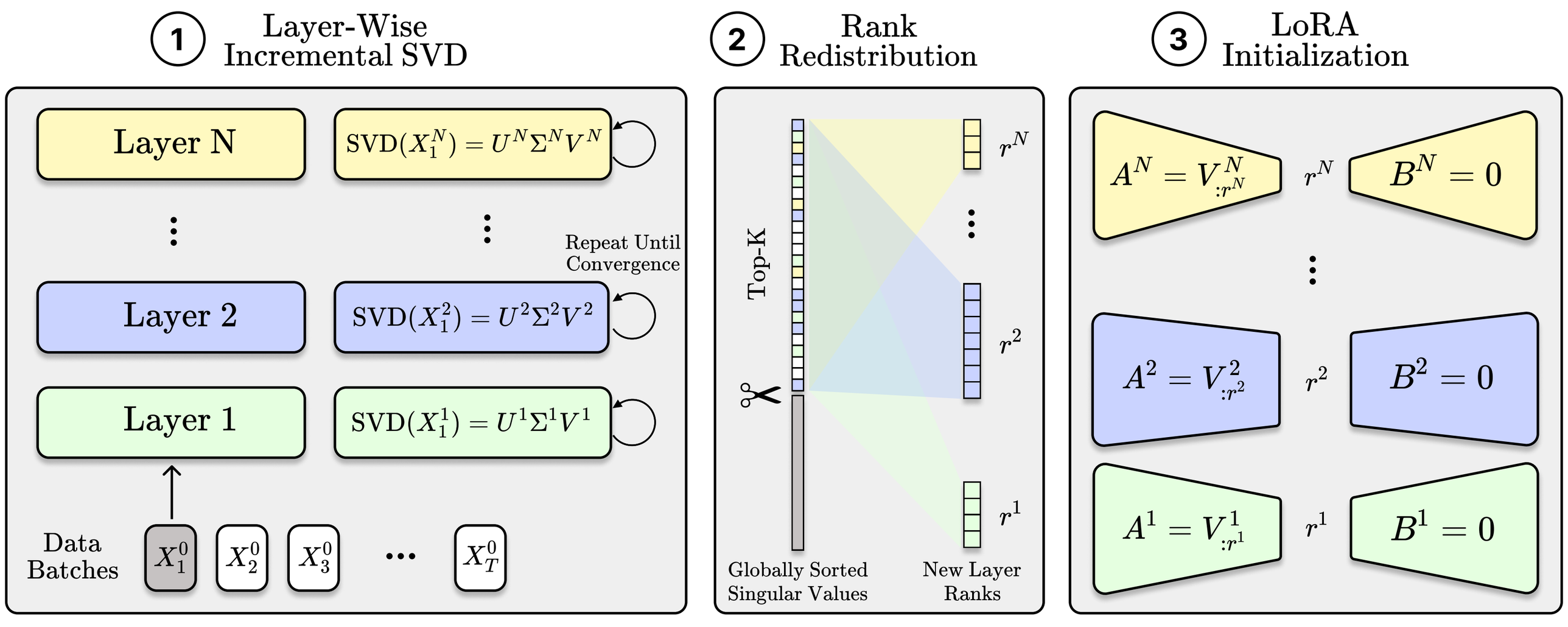

1
Layer-Wise
Incremental SVD
Layer N
Layer 2
Layer 1
$\mathrm{SVD}(X_1^N) = U^N \Sigma^N V^N$
$\mathrm{SVD}(X_1^2) = U^2 \Sigma^2 V^2$
$\mathrm{SVD}(X_1^1) = U^1 \Sigma^1 V^1$
Repeat Until
Convergence
Data
Batches
X_1^0
X_2^0
X_3^0
X_T^0
2
Rank
Redistribution
Top-K
r^N
r^2
r^1
Globally Sorted
Singular Values
New Layer
Ranks
3
LoRA
Initialization
$A^N = V^N_{:r^N}$
$B^N = 0$
$A^2 = V^2_{:r^2}$
$B^2 = 0$
$A^1 = V^1_{:r^1}$
$B^1 = 0$
r^N
r^2
r^1